Tertiary Infotech Academy Pte Ltd

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LESSON PLAN

For



Electrostatic Discharge (ESD) Control for Lab and Factory Engineers and Managers

TGS Ref No: TGS-2024051412

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 3.1

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DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
2.0	Legacy release	Legacy two-day outline.	Tertiary Infotech Academy Pte Ltd
3.0	19 August 2026	Rebuilt two-day plan aligned to the v3.0 slide map, eight activities and WA plus Case Study.	Tertiary Infotech Academy Pte Ltd
3.1	20 August 2026	Updated slide map for the logically sequenced v3.1 deck with integrated ESD photographs and contextual technical figures.	Tertiary Infotech Academy Pte Ltd

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Course Overview

Two-day WSQ programme. TSC: Electrostatic Discharge Control (ELE-OPR-4002-1.1). Assessment: Written Assessment (SAQ) — 4 open-ended questions, K1–K4, 1 hour, open book. Case Study (CS) — 2 open-ended tasks, A1–A3, 1 hour, open book.

Learning Outcomes

- LO1: Apply control measures to manage ESD generation based on ESD principles.
- LO2: Evaluate and recommend actions to improve ESD control effectiveness in accordance with ESD standards.

Day 1 Schedule

Time	Session / learning activity	Slide reference
09:00–09:30	Digital attendance, introductions and course briefing	Admin / assessment
09:30–10:30	Module 1: charge generation, discharge paths and models	15–49
10:30–10:40	Morning break	Admin / assessment
10:40–12:15	Failure modes, exposure points and business impact	15–49
12:15–13:00	Activities 1–2: hazard hunt and failure triage	15–49, A1: 43–45, A2: 46–48
13:00–13:45	Lunch	Admin / assessment
13:45–15:15	Module 2: EPA, grounding, personnel and worksurface controls	50–77
15:15–15:25	Afternoon break	Admin / assessment
15:25–16:45	Ionisation, shielding, packaging and material movement	50–77
16:45–17:45	Activities 3–4: EPA design and automated-cell controls	A3: 71–73, A4: 74–76
17:45–18:00	Day 1 recap and evidence check	Admin / assessment

Day 2 Schedule

Time	Session / learning activity	Slide reference
09:00–09:20	Digital attendance and Day 1 retrieval practice	Admin / assessment
09:20–10:30	Module 3: standards and programme architecture	79–102
10:30–10:40	Morning break	Admin / assessment
10:40–12:00	Measurement discipline, instruments and verification	79–102
12:00–13:00	Activities 5–6: verification round and standards audit	A5: 96–98, A6: 99–101
13:00–13:45	Lunch	Admin / assessment
13:45–14:45	Module 4: competence, incident	104–123

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	response and CAPA	
14:45–14:55	Afternoon break	Admin / assessment
14:55–15:35	Activity 7: incident investigation	A7: 117–119
15:35–16:00	Activity 8: management capstone and course summary	A8: 120–122
16:00–17:00	Written Assessment (SAQ) — 1 hour	Admin / assessment
17:00–18:00	Case Study (CS) — 1 hour	Admin / assessment

Activity and Assessment Alignment

Activity	K/A mapping	Output	Slides
1. Static Hazard Hunt at NovaSense Incoming Inspection	K1 · A1	Annotated hazard map; Risk-prioritisation worksheet; Three-control proposal; Peer-review note	43–45
2. Medical Microchip Failure Triage: ESD, EOS or Process Defect?	K1 · A1	Chronology; Hypothesis matrix; Containment plan; Decision-gate briefing	46–48
3. Design and Commission an EPA for Orion Optics	K3 · K4 · A1	EPA layout; Grounding/bonding diagram; Insulator-control register; Commissioning plan and readiness certificate	71–73
4. Select Controls for the Aster Robotics Handling Cell	K3 · K4 · A1	Exposure-path map; Control-selection matrix; Monitoring/stop-condition plan; Maintenance plan	74–76
5. Execute a Compliance Verification Round	K2 · K3 · A2	Completed measurement log; Failure/disposition records; Metric calculation; Signed evidence package	96–98
6. Standards-Based EPA Audit at Helix Assembly	K2 · A2 · A3	Audit plan; Completed checklist; Evidence register; Graded findings and follow-up plan	99–101
7. Investigate the PulseMed ESD Incident	K4 · A2 · A3	Containment record; Timeline and causal analysis; CAPA plan; Effectiveness-verification plan	117–119
8. Management Capstone: Build the Apex ESD Control Plan	K1–K4 · A1–A3	ESD control plan; Qualification and verification matrices; Competency matrix; Metrics dashboard and 90-day roadmap	120–122

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